



Faculty of Information Technology
Computer Science Department

KEY

Web Applications and Technologies - COMP334
Midterm Exam

Date: Sunday, 10/04/2016

Time: 80 minutes

Choose Your Instructor: ☐ Dr. Yousef A Hassouneh

☐ Dr. Samer Al-Zain

Student Number:	
Student Name:	

Instructions:

- ✓ This is a closed book exam. Answer all of the four questions, no optional questions.
- ✓ Answer the questions clearly in the space provided. Use the back of the pages for your rough work
- ✓ Students' questions are allowed only in the first 10 minutes after distribution the exam papers.
- ✓ Leaving the exam not before half the exam time being passed.

Question #	1	2	3	4	5	Total of 100
Grade						

Question # 1 (20 points) :

A. Describe the components of a typical three-tier web application.

Presentation Tier
• Viewing / Rendering html, Internet browsers, many clients

Application Tier
• Application & business related logic

Database Tier
• Database storage & operations such as read/write/
update/delete -- etc

B. Write down the steps for the process of requesting and receiving a static web page (HTML) and the process of requesting and receiving a dynamic web page (PHP). Your answer should clarify the roles of web browser, web server and php interpreter.

Static page

- 1- IB sends http request to web server
- 2- Web server process requests
- 3- Web server searches for html page
- 4- Web server returns html page through http response to browser
- 5- IB renders html to user

- ### Dynamic Web page
- 1- IB sends http request to web server
 - 2- Web server process request, and delegates request to php interpreter
 - 3- php interpreter processes php page: html is copied & php is executed
 - 4- The result is html page that is sent to IB
 - 5- IB renders HTML to user

C. What is the difference between GET and POST methods.

GET

- 1- submitted data is displayed in the url / not secure
- 2- Limited amount of data is allowed

POST

- 1- submitted data is hidden inside the http request.
- 2- no limits for data size

D. What is the difference between Block and Inline elements in HTML, give two examples on each?

Block: displayed alone on page, ^{other} elements are displayed after or before it, example <div> and <p>

Inline: displayed on same section line
example: <a> and

Question # 2 (10 points) :

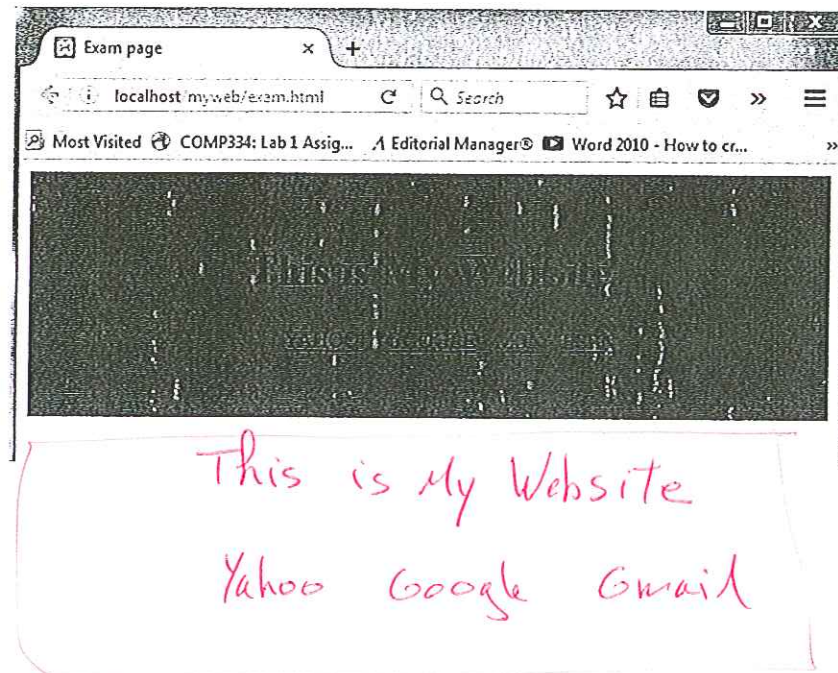
Determine whether each of the following statements are TRUE or FALSE:

T/F	Statements
T	1. The element <a> is an inline element.
F	2. Clients (browsers) Provide HTML pages to the server.
F	3. HTML syntax is much more strict than XHTML, leading to clean and clear documents in a standard form.
F	4. HTML with CSS can be used together to dynamically create pages on a web server.
F	5. The <caption> for a table should be located right before the <table> tag
T	6. The <meta> tags are processed and understood by Internet Browsers.

Question # 3 (15 points)

A. Write XHTML codes and CSS that generate the following part of a webpage :

NOTE: the links should be built using list items and anchors (, <a>). The CSS should be internal and assume that the back color is green.



```
<html>
<head> <title> Page Name </title> </head>
<style type="text/css">
  div { padding: 30px; background-color: green; text-align: center; }
  li { display: inline; margin-right: 10px; } </style> </head>
<body>
  <div> <h1> This is My Website </h1>
  <ul>
    <li> <a href="http://www.yahoo.com"> Yahoo </a> </li>
    <li> <a href="http://www.google.com"> Google </a> </li>
    <li> <a href="http://www.google.com"> Gmail </a> </li>
  </ul>
  </div>
</body>
</html>
```

Question # 4 (25 points) :

Write a suitable CSS rule for each of the following styles.

- A. All Heading at level 2 through 5 should be in center alignment.

```
h2, h3, h4, h5 {  
  text-align: center;  
}
```

- B. The paragraphs that belong to class "shades" only should be preceded by a 3 cm indentation and 2 points blue border.

```
p.shades {  
  border: 2px solid blue;  
  border-color: blue;  
  text-indent: 3cm 3cm;  
}
```

- C. Suppose that we have a `div` element with `id="d1"` and with specific height and width, we want it to be displayed exactly at x,y coordinates 100, 300.

```
#d1 {  
  position: absolute;  
  left: 100px;  
  top: 300px;  
}
```

- D. A hyperlink that is active should appear on orange and larger by 1.5 em.

```
a:active {  
  color: orange;  
  font-size: 1.5em;  
}
```

- E. All paragraphs that have the "later" class should be hidden.

```
p.later {  
  visibility: hidden;  
}
```

Question # 5 (30 points) :

Write a dynamic web page containing HTML and PHP script to do the following:

- The HTML page will contain a form that allows users to enter a player's number and then submit it to a script called "players.php", using POST method.
- The "players.php" initialize an associative array, in which the player number is the key and the goal count is the value.
- The "players.php" search the array for the player number, if exist the player exit the script should create a dynamic HTML page that displays the number of goals has been registered by the player. If the player does not exist, the script creates an HTML page to display a message that indicate no such player exist.
- The "players.php" should display a welcome message when page is first requested.

players (2)

```
<html>
<head>
<title>
Player Scores
</title>
</head>
<body>
<?php
$players = array(10=>5, 15=>8 , 5=> 6, 12=>2);

if (isset($_POST['submit']) && isset($_POST['playerID']))
{
    $id = $_POST['playerID'];
    if (isset ($players[$id]))
        echo "<h2> Players whose number is ".$id. "has registered".
$players[$id]. "scores</h2>";
    else
        echo "<h2>Sorry no Player with number".$id."has been found<h2>";
}
else
    echo "<h2>no player's id has been sent</h2>";
?>
</body>
</html>
```